


**SIDDHARTH INSTITUTE OF ENGINEERING & TECHNOLOGY:: PUTTUR
(AUTONOMOUS)**

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Subject with Code: (23CS0508) OBJECT ORIENTED PROGRAMMING THROUGH JAVA

Course & Branch: B.Tech – CSE, CSM, CAD, CIC, CCC, CAI,CIA and CSIT

Regulation: R23

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UNIT-I

1. Which of the following is an Object-Oriented Programming principle? []			
a) Inheritance	b) Recursion	c) Stack	d) None of the above
2. Which of the following is not a Java token? []			
a) Keywords	b) Operators	c) Identifiers	d) Libraries
3. How do you compile a Java program from the command line? []			
a) javac filename.java	b) java filename.java	c) compile filename.java	d) run filename.java
4. Which of the following escape sequences is used to represent a new line in Java? []			
a) \r	b) \n	c) \t	d) \\
5. Which of the following is not a primitive data type in Java? []			
a) int	b) String	c) float	d) boolean
6. Which of the following is a valid declaration of a variable in Java? []			
a) int 2x=10;	b) int x=10;	c) int \$y=10;	d) int -z=10;
7. What does the term "type casting" refer to in Java? []			
a) Converting data from one type to another	b) Creating objects	c) Defining constants	d) Passing arguments to a method
8. What will be the output of System.out.printf("%d", 10);? []			
a) 10	b) %10	c)IllegalFormatException	d) d
9. Which keyword is used to define a constant in Java? []			
a) static	b) final	c) constant	d) const
10. What will be the output of System.out.println('A' + 1);? []			
a) A1	b) B	c) 66	d) 65
11. What is the output of the following code? [] int a = 5, b = 10;System.out.println(a < b ? a : b);			
a) 5	b) 10	c) true	d) false
12. What is the output of System.out.println(10 % 3);? []			
a) 3	b) 0	c) 1	d) 2
13. What is the default value of a static variable in Java? []			
a) null	b) 0	c) undefined	d) Depends on the data type
14. What does the keyword static indicate in Java? []			

a) The variable or method belongs to the class, not an instance	b) The variable or method belongs to an instance	c) The variable or method can be overridden	d) The variable or method can only be used once
15. Which of the following is a valid method declaration in Java? []			
a) public void method1[] { }	b) public void method1() { }	c) public method1() void { }	d) void public method1() { }
16. Which of the following operators has the highest precedence? []			
a) *	b) +	c) =	d) &&
17. What does the ++ operator do in Java? []			
a) Increases the value of the variable by 2	b) Increases the value of the variable by 1	c) Adds two values	d) None of the above
18. Which operator is known as the ternary operator? []			
a) ?:	b) &&	c)	d) %
19. Which of these is a valid ternary operator syntax? []			
a) condition ? expr1 : expr2	b) condition : expr1 ? expr2	c) condition ? expr1 , expr2	d) condition , expr1 : expr2
20. What is the output of System.out.println(10 > 5 ? "Yes" : "No");? []			
a) Yes	b) No	c) 10	d) 5
21. Which operator is used for bitwise AND operation in Java? []			
a) &&	b) &	c)	d) ^
22. Which of the following operators is used for bitwise OR in Java? []			
a) &&	b)	c)	d) ^
23. Which of the following is a relational operator in Java? []			
a) +	b) <	c) =	d) *
24. What is the default value of a boolean variable in Java? []			
a) true	b) false	c) 0	d) null
25. Which of the following is the correct syntax for an if-else statement in Java? []			
a) if (condition) { } else { }	b) if condition { } else { }	c) if (condition) else { }	d) if (condition) { } else if { }
26. How is the switch statement evaluated in Java? []			
a) Based on the value of a boolean expression	b) Based on the value of a string expression	c) Based on the value of an integer or character	d) None of the above
27. What is the result of the expression true && false? []			
a) true	b) false	c) null	d) undefined
28. What is the result of the expression 5 + 2 * 3 in Java? []			
a) 11	b) 21	c) 15	d) 7
29. Which loop will always execute at least once in Java? []			
a) while loop	b) for loop	c) do-while loop	d) All of the above
30. What does the break statement do in a loop? []			
a) Skips to the next iteration	b) Terminates the loop	c) Exits the method	d) Throws an exception
31. What does the final keyword in Java signify? []			

a) The variable cannot be changed	b) The method cannot be overridden	c) The class cannot be extended	d) All of the above
32. How do you declare an array in Java? []			
a) int[] arr = new int[5];	b) array int arr[5];	c) int arr[5];	d) int arr() = new int[5];
33. Which of the following is used to start the execution of a Java program? []			
a) static method	b) main method	c) constructor	d) public method
34. Which access modifier allows visibility within the same package only? []			
a) public	b) private	c) protected	d) No modifier (default)
35. Which of the following is the correct way to declare a symbolic constant in Java? []			
a) const int MAX = 100;	b) final int MAX = 100;	c) final MAX = 100;	d) constant int MAX = 100;
36. How do you convert an integer to a string in Java? []			
a) String.valueOf(int)	b) Integer.toString(int)	c) Both a and b	d) None of the above
37. What is the scope of a variable defined inside a method? []			
a) Global	b) Method-level	c) Class-level	d) Package-level
38. Which of the following is not a control statement in Java? []			
a) if	b) switch	c) break	d) try
39. How do you define a method that does not return any value? []			
a) void methodName()	b) int methodName()	c) return methodName()	d) empty methodName()
40. What does the continue statement do in a loop? []			
a) Stops the loop	b) Restarts the loop from the beginning	c) Skips the rest of the current iteration	d) Exits the method

UNIT-II

1. What are the four types of variables in Java? []			
a) Local, instance, static, and parameter	b) Local, instance, static, and global	c) Instance, static, parameter, and class	d) Local, instance, class, and static
2. What is the difference between instance and static variables? []			
a) Accessibility	b) Lifetime	c) Scope	d) Memory
3. Can a static variable be accessed using an instance reference? []			
a) Yes	b) No	c) Only if it's final	d) Only if it's public
4. What is the default value of an uninitialized instance variable of type int? []			
a) 0	b) null	c) undefined	d) garbage value
5. What are the two types of methods in Java? []			
a) Instance and static	b) Abstract and concrete	c) Public and private	d) Synchronized and non-synchronized
6. Can a static method access instance variables? []			
a) Yes	b) No	c) Only if they're final	d) Only if they're public
7. What happens when you modify a static variable in java? []			
a) Affect all instances	b) Affect only current instances	c) Throws exception	d) No effect
8. What is the difference between = and clone() in Java? []			
a) = creates a deep copy, clone() creates a shallow copy	b) = creates a shallow copy, clone() creates a deep copy	c) = creates a new object, clone() creates a reference	d) = creates a reference, clone() creates a new object
9. Can assigning one object to another affect the original object? []			
a) Yes	b) No	c) Only if it's final	d) Only if it's immutable
10. What is the purpose of the Cloneable interface in Java? []			
a) To enable deep copying of objects	b) To enable shallow copying of objects	c) To prevent object cloning	d) To ensure object immutability
11. What is the difference between shallow and deep copying in Java? []			
a) Syntax	b) Speed	c) Independence	d) Memory
12. How can you implement deep copying in Java? []			
a) Using clone() method	b) Using serialization	c) Using copy constructor	d) All of the above
13. Can deep copying throw exceptions in Java? []			
a) Yes	b) No	c) Only if object is final	d) Only if object is immutable
14. What is the purpose of the Cloneable interface in Java? []			
a) To enable deep copying of objects	b) To enable shallow copying of objects	c) To prevent object cloning	d) To ensure object immutability
15. What interface must be implemented for object.clone() to work? []			
a) Cloneable	b) Serializable	c) Comparable	d) Iterable
16. How can you implement deep copying in Java? []			

a) Using clone() method	b) Using serialization	c) Using copy constructor	d) All of the above
17. Can deep copying throw exceptions ? []			
a) Yes	b) No	c) Only if object is final	d) Only if object is immutable
18. Which of the following is true about a final class in Java? []			
a) can have a subclass.	b) can have constructors	c) cannot have constructors	d) must be static.
19. What is the purpose of the this keyword in Java? []			
a) To refer to the current instance	b) To refer to the current class	c) To refer to the current method	d) To refer to the current package
20. What happens when a subclass overrides a method from its superclass in Java? []			
a) Hide	b) Override	c) Extends	d) implements
21. What is the difference between method overloading and overriding? []			
a) Signatures	b) Return	c) Parameters	d) Inheritance
22. Can a subclass override a private method from its superclass? []			
a) Yes	b) No	c) Only if it's final	d) Only if it's public
23. What is the purpose of the Serializable interface in Java? []			
a) To enable object serialization	b) To enable object deserialization	c) To enable deep copying	d) To ensure object immutability
24. What is the primary requirement for method overloading? []			
a) Name	b) Return	c) Parameters	d) Class
25. What is the benefit of method overloading? []			
a) Encapsulation	b) Polymorphism	c) Abstraction	d) Readability
26. What is the term for calling a method from within the same method? []			
a) Inheritance	b) Invocation	c) Recursion	d) Iteration
27. Can method overloading be done with varying number of parameters? []			
a) Yes	b) No	c) Only with varargs	d) Only with arrays
28. What happens if two overloaded methods have same parameter list? []			
a) Compiler error	b) Runtime error	c) Method ambiguity	d) Method overriding
29. Can overloaded methods have different access modifiers? []			
a) Yes	b) No	c) Only public and private	d) Only protected and default
30. Where can static nested classes be accessed from? []			
a) Everywhere	b) inner class	c) outer class	d) only main class
31. How does Java handle method overloading with var args? []			
a) Treats var args as array	b) Treats var args as single parameter	c) Ignores var args	d) Throws compiler error
32. Can method overloading be used with static and instance methods? []			
a) Yes	b) No	c) Only with static methods	d) Only with instance methods
33. Where are local inner classes defined? []			

a) Static block	b) Methods	c) Super class	d) Sub class
34. Can abstract classes overload methods? []			
a) Yes	b) No	c) Only with abstract methods	d) Only with concrete methods
35. What is the term used to describe a method that calls itself? []			
a) Iterative	b) Recursive	c) Dynamic	d) Static
36. What is the type of invocation used in recursive methods? []			
a) Direct	b) Indirect	c) Self	d) Mutual
37. What is the condition that stops a recursive method from calling itself? []			
a) Terminal	b) Initial	c) Base	d) Recursive
38. What is the primary purpose of anonymous inner classes? []			
a) Access methods	b) Access parameters	c) Access variables	d) Implementation
39. Can a final method be overridden in a subclass? []			
a) Yes	b) No	c) Only in abstract classes	d) Only in interfaces
40. Which of the following can be declared as final in Java? []			
a) Class	b) Method	c) Variable	d) All of the above

UNIT - 3

1. How do you declare a one-dimensional array in Java? []			
a) int[] arr;	b) int arr[];	c) int arr();	d) int arr[5];
2. What is the index range for a 1-dimensional array in Java? []			
a) 0 to length-1	b) 1 to length	c) -1 to length	d) 0 to length
3. How do you access an element in a 1-dimensional array? []			
a) arr[index]	b) arr(index)	c) arr.get(index)	d) arr.set(index, value)
4. What is the purpose of the length property in a 1-dimensional array? []			
a) To get the size of the array	b) To set the size of the array	c) To access an element in the array	d) To sort the array
5. Can you change the size of a 1-dimensional array after declaration? []			
a) Yes	b) No	c) Only using Arrays.copyOf()	d) Only using System.arraycopy()
6. How do you iterate through a 1-dimensional array? []			
a) Using for loop	b) Using while loop	c) Using foreach loop	d) All of the above
7. _____ is the correct way to declare an array in Java. []			
a) int arr[] = new int[5];	b) int arr = new int(5);	c) int arr[5];	d) int arr = new int[];
8. Can you declare an array without specifying its size? []			
a) Yes	b) No	c) Only for primitive types	d) Only for object types
9. How do you declare a multi-dimensional array in Java? []			
a) int[][] arr;	b) int[] arr[];	c) int arr[][];	d) All of the above
10. What is the purpose of the new keyword in array declaration? []			
a) To allocate memory for the array	b) To initialize the array	c) To declare the array	d) To access the array
11. How do you initialize an array in Java? []			
a) int[] arr = {1, 2, 3};	b) int[] arr = new int[] {1, 2, 3};	c) int[] arr = new int[3]; arr[0] = 1; arr[1] = 2; arr[2] = 3;	d) All of the above
12. _____ is the default value of an element in an integer array in Java. []			
a) Null	b) 0	c) Undefined	d) Garbage value
13. Can you initialize an array with different data types? []			
a) Yes	b) No	c) Only for object arrays	d) Only for primitive arrays
14. What is the output of the following code? int[] arr = {1, 2, 3, 4}; System.out.println(arr[2]); []			
a) 1	b) 2	c) 3	d) 4
15. What is the process of assigning one array to another array in Java? []			
a) Copying	b) Cloning	c) Referencing	d) Assigning

16.What is the data structure in Java that allows dynamic change of size? []			
a) Array	b) List	c) Stack	d) Queue
17.____method is used to sort an array in Java. []			
a) Collections.sort()	b) Arrays.sort()	c) array.sort()	d) System.sort()
18. What is the index range for a 2-dimensional array in Java? []			
a) 0 to rows-1, 0 to columns-1	b) 1 to rows, 1 to columns	c) -1 to rows, -1 to columns	d) 0 to rows, 0 to columns
19. How do you declare a 2-dimensional array in Java? []			
a) int[][] arr;	b) int[] arr[];	c) int arr[][];	d) All of the above
20. What is the purpose of the length property in a 2-dimensional array? []			
a) To get the number of rows	b) To get the number of columns	c) To get the total number of elements	d) To get the row and column sizes
21. How do you iterate through a 2-dimensional array? []			
a) Using nested for loops	b) Using nested while loops	c) Using foreach loop	d) All of the above
22. Can you have jagged arrays in Java? []			
a) Yes	b) No	c) Only for primitive types	d) Only for object types
23. Does Java support multiple inheritance? []			
a) Yes, using classes	b) No, but using interfaces	c) Yes, using interfaces and classes	d) No, Java does not support multiple inheritance
24. Why does Java not support multiple class inheritance? []			
a) To avoid ambiguity	b) To improve performance	c) To reduce complexity	d) To increase security
25. How can you achieve multiple inheritance in Java? []			
a) Using interfaces	b) Using abstract classes	c) Using composition	d) All of the above
26. What is the diamond problem in multiple inheritance? []			
a) Ambiguity in method overriding	b) Ambiguity in method overloading	c) Ambiguity in interface implementation	d) Ambiguity in class inheritance
27. Can a class implement multiple interfaces in Java? []			
a) Yes	b) No	c) Only if interfaces are related	d) Only if interfaces are unrelated
28. What is the purpose of implementing multiple interfaces? []			
a) To achieve multiple inheritance	b) To increase code complexity	c) To decrease code reusability	d) To improve code readability
29. How do you implement multiple interfaces in Java? []			
a) Using implements keyword separated by commas	b) Using extends keyword separated by commas	c) Using implements keyword multiple times	d) Using extends keyword multiple times
30. Can multiple interfaces have same method signature? []			
a) No	b) Yes	c) Only if methods are static	d) Only if methods are instance

31. How does Java handle method ambiguity in multiple interfaces? []			
a) Method overriding	b) Runtime error	c) Compiler Error	d) Method overloading
32. What is a nested interface in Java? []			
a) Interface inside another interface	b) Interface inside a class	c) Class inside an interface	d) Class inside another class
33. Why are nested interfaces used? []			
a) To group related interfaces	b) To hide interface implementation	c) To improve code readability	d) To reduce code complexity
34. Can nested interfaces be public? []			
a) Yes	b) No	c) Only if outer interface is public	d) Only if outer class is public
35. How do you access a nested interface? []			
a) Using outer interface/class name	b) Using nested interface name directly	c) Using outer interface/class name and nested interface name	d) Using import statement
36. Can nested interfaces extend other interfaces? []			
a) Yes	b) No	c) Only if outer interface extends	d) Only if outer class extends
37. How does Java resolve multiple inheritance conflicts? []			
a) Using method overriding	b) Using method overloading	c) Using interface implementation	d) Using class hierarchy
38. What is the purpose of the super keyword in Java? []			
a) To access superclass members	b) To access subclass members	c) To override method	d) To overload methods
39. How do you use super to access superclass constructors? []			
a) super()	b) super.constructor()	c) this.super()	d) super(this)
40. Can super be used in static methods? []			
a) Yes	b) No	c) Only in subclass	d) Only in superclass

UNIT - 4

1. What is the purpose of packages in Java? []			
a) To organize classes and interfaces	b) To provide security	c) To improve performance	d) To reduce complexity
2. How do you declare a package in Java? []			
a) Using package keyword	b) Using import keyword	c) Using class keyword	d) Using interface keyword
3. What is the default package in Java? []			
a) java.lang	b) java.util	c) default	d) unnamed package (no explicit package declaration)
4. Can classes in different packages have same name? []			
a) Yes	b) No	c) Only if classes are abstract	d) Only if classes are final
5. How do you access a class from another package? []			
a) Using import statement	b) Using fully qualified name	c) Using package name	d) Using class name
6. What is the purpose of import statement in Java? []			
a) To declare a package	b) To access classes from another package	c) To provide security	d) To improve performance
7. What is the difference between import and fully qualified name? []			
a) import is used for packages, fully qualified name is used for classes	b) import is used for classes, fully qualified name is used for packages	c) import is used for static imports, fully qualified name is used for dynamic imports	d) import is used for dynamic imports, fully qualified name is used for static imports
8. What is the environment variable that specifies the directory path for Java classes?			
a) PATH	b) CLASSPATH	c) JAVAHOME	d) JREHOME
9. What is the environment variable that specifies the directory path for executable files?			
a) PATH	b) CLASSPATH	c) JAVAHOME	d) JREHOME
10. What is package-private access? []			
a) Accessible only within same package	b) Accessible only within same class	c) Accessible only within same method	d) Accessible from anywhere
11. Can protected members be accessed from another package? []			
a) Yes, using inheritance	b) No	c) Yes, using import statement	d) Yes, using fully qualified name
12. What is the keyword used to declare a package in Java? []			
a) Import	b) Package	c) Class	d) Public
13. What is the package that is automatically imported in every Java program?			
a) Java	b) Lang	c) Util	d) Io
14. What is an instance of a class in Java?			

a) Class	b) Object	c) Instance	d) Variable
15.What is the keyword used to declare an enumeration in Java?			
a) Class	b) Interface	c) Enum	d) Abstract
16.What is the class in Java that provides mathematical functions?			
A) Math	B) Arithmetic	C) Calculator	D) Geometry
17.What are the classes in Java that wrap primitive data types?			
a) Primitive	b) Wrapper	c) Container	d) Boxed
18.What is the process of automatically converting a primitive data type to its corresponding wrapper class in Java?			
a) Autoboxing	b) Auto-unboxing	c) Wrapping	d) Boxing
19.What is the process of automatically converting a wrapper class to its corresponding primitive data type in Java?			
a) Autoboxing	b) Auto-unboxing	c) Unwrapping	d) Unboxing
20.What is the package in Java that provides utility classes for collections, dates, and other utilities?			
a) Java.lang	c) Java.util	c)java.io	d)java.time
21.What is the class in Java that provides methods for formatting output?			
a) Formatter	b) Scanner	c) PrintWriter	d) BufferedReader
22.What is the class in Java that provides methods for generating random numbers?			
A) Random	B) Math	C) Scanner	D) Formatter
23. What is the purpose of the java.time package? []			
a) To provide date and time utilities	b) To provide calendar utilities	c) To provide scheduling utilities	d) To provide timezone utilities
24. Which class represents the current date and time in java.time? []			
a) LocalDate	b) LocalTime	c) LocalDateTime	d) Instant
25. What is the difference between LocalDate and LocalDateTime? []			
a) LocalDate includes time, LocalDateTime excludes time	b) LocalDate excludes time, LocalDateTime includes time	c) LocalDate is for UTC, LocalDateTime is for timezone	d) LocalDate is for timezone, LocalDateTime is for UTC
26. How do you get the current timezone in java.time? []			
a) ZoneId.systemDefault()	b) ZoneId.of("UTC")	c)TimeZone.getDefault()	d)Calendar.getInstance().getTimeZone()
27. How do you convert LocalDateTime to Instant? []			
a) localDateTime.atZone(ZoneId.systemDefault())	b)localDateTime.atZone(ZoneId.of("UTC"))	c)localDateTime.toInstant()	d)Instant.from(localDateTime)
28. What is the purpose of Period class in java.time? []			
a) To represent date intervals	b) To represent time intervals	c) To represent timezone offsets	d) To represent calendar periods
29.Which of the following is a checked exception in Java? []			
a)NullPointerException	b) IOException	c)ArrayIndexOutOfBoundsException	d) RuntimeException
30.What type of exception in Java is not checked at compile-time?			

a) Checked	b) Unchecked	c) Runtime	d) Error
31. What is the difference between throw and throws in Java? []			
A) throw is used for checked exceptions, throws for unchecked	B) throw is used to declare exceptions, throws to throw exceptions	C) throw is used to throw exceptions, throws to declare exceptions	D) throw is used for runtime exceptions, throws for compile-time exceptions
32. What is the block of code in Java that is used to enclose code that might throw an exception?			
a) Try	b) Catch	c) Finally	d) Throw
33. What is the block of code in Java that is used to handle an exception that is thrown by the try block?			
a) Try	b) Catch	c) Finally	d) Throw
34. What is the purpose of the try-catch-finally block in Java? []			
A) To handle errors only	B) To handle exceptions and release resources	C) To handle exceptions only	D) To skip exception handling
35. What is the feature in Java that allows multiple exceptions to be caught and handled by a single catch block?			
a) Multi-catch	b) Try-catch	c) Exception handling	d) Error handling
27. What is the difference between Exception and RuntimeException? []			
A) Exception is checked, RuntimeException is unchecked	B) Exception is unchecked, RuntimeException is checked	C) Exception is runtime, RuntimeException is compile-time	D) Exception is compile-time, RuntimeException is runtime
28. Can you throw a checked exception from a method that is not declared to throw it? []			
A) Yes	B) No	C) Only if it's subclass of RuntimeException	D) Only if it's subclass of Exception
29. What happens if an exception is thrown in a method and not caught? []			
A) Program terminates	B) Exception is propagated to caller method	C) Exception is ignored	D) Method returns null
30. How do you create a custom exception in Java? []			
A) Extending Exception class	B) Implementing Exception interface	C) Extending RuntimeException class	D) Implementing RuntimeException interface
32. Which of the following exceptions is an example of an unchecked exception in Java? []			
A) IOException	B) NullPointerException	C) SQLException	D) FileNotFoundException
33. What is the purpose of multiple catch blocks in Java? []			
A) To handle different types of exceptions	B) To handle different types of errors	C) To improve code readability	D) To reduce code length
35. How can you catch multiple exceptions in a single catch block? []			
A) Using catch (Exception1 Exception2)	B) Using catch (Exception1, Exception2)	C) Using catch (Exception1 && Exception2)	D) Using catch (Exception1 Exception2)
36. What is the standard input stream in Java? []			
A) System	B) Input	C) Output	D) Error

37.What type of streams in Java are used to read and write raw bytes? []			
A) Byte	B) Character	C) Input	D) Output
38.What type of streams in Java are used to read and write characters? []			
A) Byte	B) Character	C) Input	D) Output
39.What is the class in Java used to get input from the user or a file. []			
A) Reader	B) Scanner	C) Writer	D) Stream
40.What is the class in Java used to handle file operations? []			
A) File	B) Files	C) Reader	D) Writer

UNIT-5

1. What is the purpose of the CharSequence interface in Java? []			
A) To represent a mutable character sequence	B) To represent an immutable character sequence	C) To provide a common interface for string manipulation	D) To provide a way to read character streams
2. Which of the following classes implements CharSequence? []			
A) String	B) StringBuilder	C) StringBuffer	D) All of the above
3. What method is declared in the CharSequence interface to extract a subset of characters? []			
A) substring()	B) subSequence()	C) charAt()	D) length()
4. What is the purpose of the String class in Java? []			
A) To represent a mutable character sequence	B) To represent an immutable character sequence	C) To provide input/output operations	D) To perform mathematical calculations
5. Which method is used to concatenate two strings in Java? []			
A) +	B) concat()	C) join()	D) append()
6. What is the difference between String and StringBuilder? []			
A) String is mutable, StringBuilder is immutable	B) String is immutable, StringBuilder is mutable	C) String is for input, StringBuilder is for output	D) String is for output, StringBuilder is for input
7. Which method is used to extract a substring from a string? []			
A) substring()	B) subSequence()	C) split()	D) charAt()
8. How do you compare two strings in Java? []			
A) Using ==	B) Using .equals()	C) Using .compareTo()	D) Using .equalsIgnoreCase()
9. What is the purpose of the trim() method? []			
A) To remove whitespace from the beginning and end of a string	B) To remove whitespace from the middle of a string	C) To add whitespace to a string	D) To convert a string to uppercase
10. Which method is used to split a string into an array of substrings? []			
A) split()	B) substring()	C) subSequence()	D) charAt()
11. How do you convert a string to uppercase in Java? []			
A) Using toUpperCase()	B) Using toLowerCase()	C) Using capitalize()	D) Using toTitleCase()
12. What is the purpose of the replace() method? []			
A) To replace a substring with another string	B) To remove a substring from a string	C) To add a substring to a string	D) To split a string into an array
13. What is the method in Java used to search for a substring in a string? []			
A) Index	B) Searc	C) Find	D) indexOf
14. What is the class in Java used to represent a mutable string? []			
A) String	B) StringBuffer	C) StringBuilder	D) StringTokenizer
15. What is the purpose of the Thread class in Java? []			

A) To create a new process	B) To create a new thread	C) To synchronize threads	D) To communicate between threads
16. How do you create a daemon thread in Java? []			
A) Using setDaemon() method	B) Using isDaemon() method	C) Using daemon() method	D) Using run() method
17. Which method is used to start a thread in Java? []			
A) start()	B) run()	C) init()	D) resume()
18. What is the difference between sleep() and wait()? []			
A) sleep() is for threads, wait() is for processes	B) sleep() is for processes, wait() is for threads	C) sleep() releases lock, wait() doesn't	D) sleep() doesn't release lock, wait() does
19. What is the purpose of the join() method in Java? []			
A) To wait for a thread to finish	B) To start a new thread	C) To synchronize threads	D) To communicate between threads
20. What is the difference between Thread and Runnable? []			
A) Thread is a class, Runnable is an interface	B) Thread is an interface, Runnable is a class	C) Thread is used for multithreading, Runnable for single threading	D) Thread is for single threading, Runnable for multithreading
21. What is the purpose of the synchronized keyword in Java? []			
A) To ensure thread safety	B) To improve performance	C) To reduce memory usage	D) To handle exceptions
22. Which of the following methods can be used to synchronize a block of code? []			
A) synchronized block	B) lock() method	C) semaphore() method	D) atomic() method
23. What is the difference between synchronized method and synchronized block? []			
A) Method-level synchronization vs. block-level synchronization	B) Class-level synchronization vs. object-level synchronization	C) Thread-level synchronization vs. process-level synchronization	D) None of the above
24. Which class is used to manually lock and unlock synchronization in Java? []			
A) Monitor	B) Semaphore	C) Mutex	D) Lock
25. What is a deadlock in multithreading? []			
A) When two threads are blocked indefinitely	B) When two threads are running concurrently	C) When a thread is terminated abruptly	D) When a thread is waiting for another thread to finish
26. What is the purpose of the Lock interface in Java? []			
A) To synchronize threads	B) To communicate between threads	C) To provide exclusive access to resources	D) To handle thread interruptions
27. Which class is used for thread scheduling in Java? []			
A) Thread	B) Runnable	C) Scheduler	D) Executor
28. How do you handle thread priority in Java? []			
A) Using setPriority() method	B) Using getPriority() method	C) Using priority() method	D) Using run() method
29. What is the difference between Executor and ExecutorService? []			

A) Executor is for single threading, ExecutorService for multithreading	B) Executor is for multithreading, ExecutorService for single threading	C) Executor is an interface, ExecutorService is a class	D) Executor is a class, ExecutorService is an interface
30. Which method is used to shut down an executor in Java? []			
A) shutdown()	B) shutdownNow()	C) terminate()	D) stop()
31. Which class is used for thread communication in Java? []			
A) Thread	B) Runnable	C) wait-notify	D) Semaphore
32. How do you handle thread interruptions in Java? []			
A) Using interrupt() method	B) Using stop() method	C) Using suspend() method	D) Using resume() method
33.What is the method in Java used to suspend a thread? []			
A) start	B) run	C) suspend	D) resume
34.What is the method in Java used to resume a thread? []			
A) start	B) run	C) suspend	D) resume
35.What is the method in Java used to stop a thread? []			
A) stop	B) suspend	C) resume	D) destroy
36.What is the primary purpose of JDBC? []			
A) Networking	B) Database	C) Security	D) Interface
37.What is the name of the popular open-source database management system used in JDBC tutorials? []			
A) MySQL	B) Oracle	C) PostgreSQL	D) Microsoft
38.What is the name of the JDBC driver for MySQL? []			
A) MySQL Connector/J	B) MySQL Connector/ODBC	C) MySQL Driver	D) JDBC Driver
39.What is the tool used to design and build JavaFX user interfaces []			
A) Eclipse	B) NetBeans	C) SceneBuilder	D) IntelliJ
40.What is the JavaFX class used to handle events? []			
a) EventHandler	B) EventListener	C) Handler	D) Listener